

Basilisk MPI CPU Benchmarks for Multiphase Flow

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Abstract

Basilisk is an open-source framework for solving interfacial flows on quadtree and octree meshes. Its pressure projection places a multigrid Poisson solve inside every incompressible time step, making communication across mesh levels a direct measure of the strong-scaling limit. This report sets out a portable CPU benchmark suite built from two stock Basilisk kernels and one representative two-phase calculation. Measurements on MareNostrum 5 and Snellius cover up to 16,384 MPI ranks for fixed-size kernel problems and up to 1,024 ranks for the application cases. The kernel data support small-to-medium system qualification and expose the onset of communication-limited scaling. Tests above 100 fully populated 192-core nodes remain a proposed campaign rather than a demonstrated result.

1 What the benchmark measures

An incompressible Basilisk calculation advances advection, viscosity and interfacial forces before projecting the velocity field onto a divergence-free space. The projection requires a multigrid Poisson solve. Restriction and prolongation transfer information between mesh levels, while the domain decomposition exchanges data between MPI ranks. At sufficiently large rank counts, this communication ceases to be hidden by the work performed on each rank. A fixed-size rank sweep therefore identifies both the useful strong-scaling range and the point at which adding CPUs no longer reduces the time to solution.

The proposed suite has three layers:

1. stock Poisson and Laplacian kernels for reproducible machine-to-machine comparison;
2. a verified two-phase migration problem that exercises the same solver stack with interface reconstruction and surface tension; and
3. a workload series in which spatial resolution or the number of drops is increased, testing whether a larger application postpones strong-scaling saturation.

The source cases, compact timing tables and plotting scripts are maintained at <https://github.com/comphy-lab/hpc-basilisk-scaling>. The stock kernel sources are retained without modification from the corresponding Basilisk tests [1, 2].

2 Kernel benchmark evidence

The two-dimensional kernel uses a full quadtree at level $L = 14$, corresponding to 2^{28} cells. Figure 1 compares wall-clock and MPI time on MareNostrum 5 and Snellius with the published Curie measurements. On Snellius, the Poisson time continues to decrease through 8,192 ranks and is unchanged within the resolution of this single-run dataset at 16,384 ranks. The simultaneous rise in the communication fraction identifies the large-rank regime as communication limited rather than compute limited.

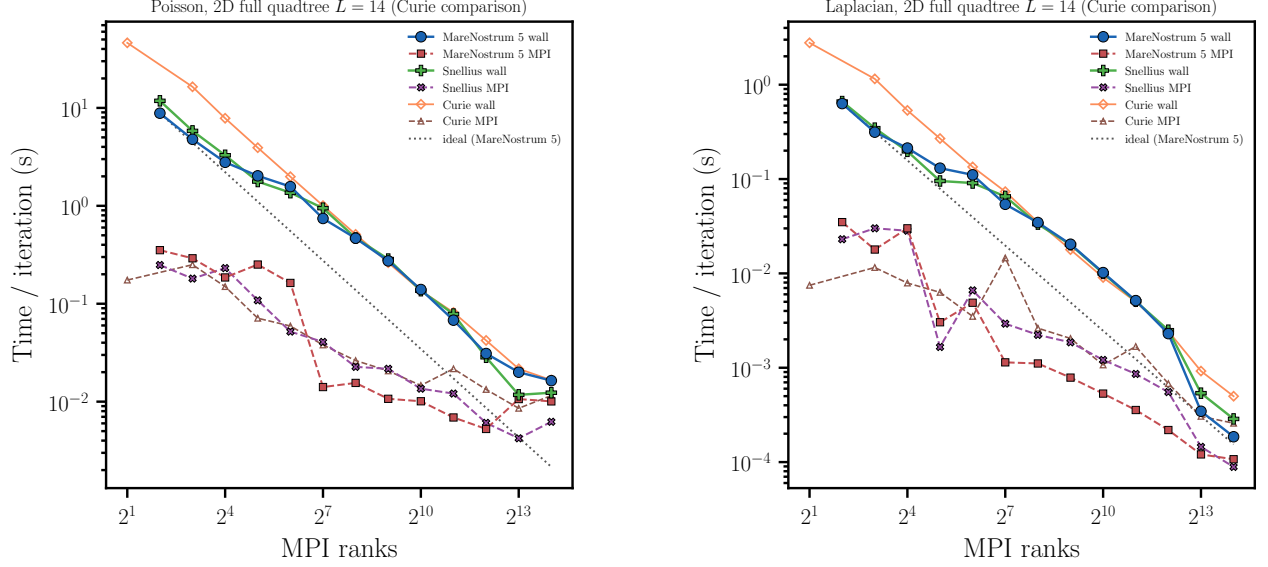


Figure 1: Strong scaling of the Poisson solve (left) and Laplacian operator (right) for the two-dimensional full-quadtree kernel at $L = 14$. Solid curves show wall time per iteration and dashed curves show the measured MPI contribution. The grey dotted line is ideal scaling from the MareNostrum 5 baseline. Curie data are the reference measurements distributed with the Basilisk test.

The three-dimensional kernel uses a full octree at $L = 9$, or 2^{27} cells. On Snellius, the Poisson time falls from 19.34 s at two ranks to 0.0333 s at 16,384 ranks. Most of the reduction has already been obtained by 4,096 ranks, where the corresponding time is 0.0390 s. Figure 2 therefore supplies both a broad scaling curve and a clear diagnostic of the strong-scaling limit. The Occigen measurements provide an independent reference for the same stock calculation.

These results show why one number is insufficient for procurement. A low-rank timing measures core and memory performance, while the bend in the curve measures the communication stack and the implementation of collective operations. Both are relevant to production multiphase-flow calculations.

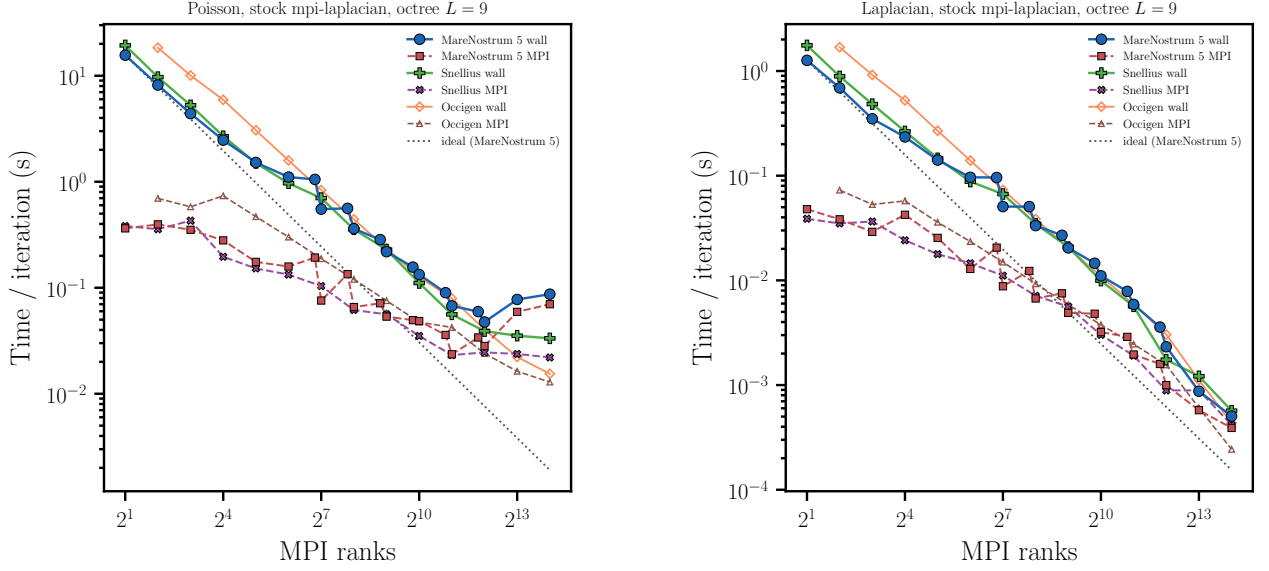


Figure 2: Strong scaling of the stock three-dimensional `mpi-laplacian` test on a full octree at $L = 9$. Symbols and line styles follow Figure 1; the reference machine is Occigen. The flattening of the Poisson curve beyond a few thousand ranks is the expected fixed-problem communication limit, not a loss of numerical correctness.

3 Representative multiphase workload

The application benchmark follows a drop driven by a surface-tension gradient. The numerical formulation is the conservative, balanced treatment of interfacial stresses described by Abu-Al-Saud, Popinet and Tchepi [4]. The cluster case adds timing, restart and controlled output to the public Basilisk test without changing the physical validation target.

Figure 3 varies the amount of work in two ways. Panel (a) increases the spatial resolution of an axisymmetric uniform mesh. Panel (b) increases the number of drops in a planar domain while retaining 64 points per radius. The larger problems continue to benefit from additional ranks after the smaller problems have saturated. This is the expected application behaviour: the useful rank count grows with the work available per rank. The present application dataset stops at 1,024 ranks, approximately 5.3 fully populated 192-core nodes, and should be used as small-scale evidence.

The application remains useful only if it solves the intended flow. Figure 4 compares the computed drop speed with the analytical terminal velocity and with the Basilisk reference data. The coarse-resolution error follows the expected second-order trend through 32 points per radius; the finer results approach the theoretical speed but should not be described as a single second-order convergence range because their error plateaus.

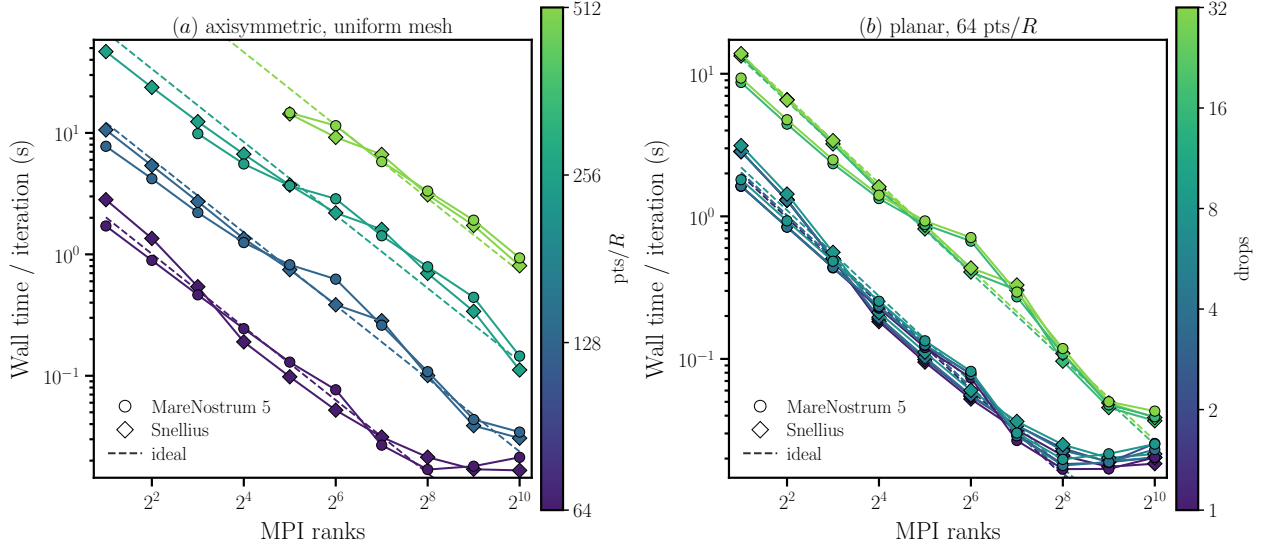


Figure 3: Wall time per iteration for representative two-phase workloads on MareNostrum 5 and Snellius. (a) Axisymmetric uniform meshes, coloured by points per drop radius. (b) Planar uniform meshes at 64 points per radius, coloured by the number of drops. Dashed lines show ideal scaling for each workload. Each point is one recorded run; the current tables do not measure run-to-run variability.

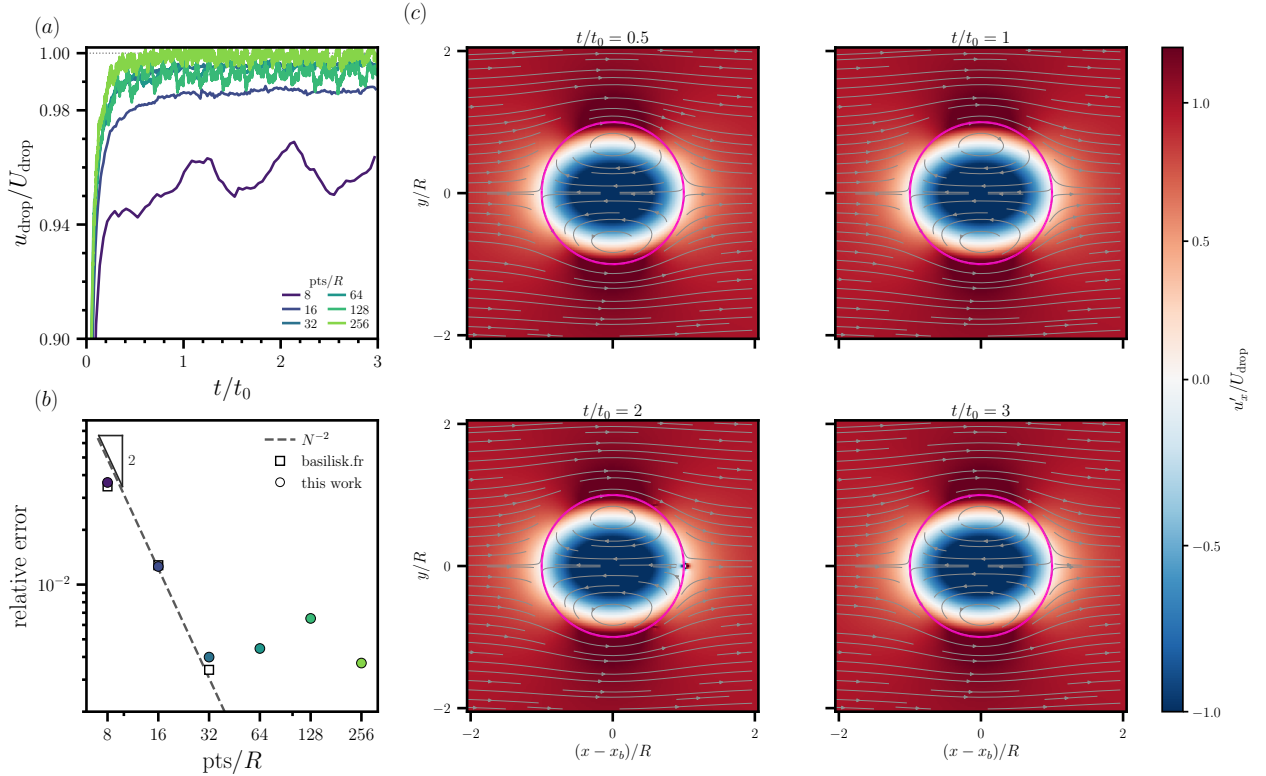


Figure 4: Numerical validation of the representative two-phase case. (a) Drop speed normalized by the theoretical terminal velocity. (b) Relative error with the N^{-2} coarse-grid trend and the public Basilisk reference. (c) Resolved velocity fields at four times. The magenta curve marks the interface and the colour scale gives the streamwise velocity normalized by the terminal speed.

4 Qualification matrix for a 192-core node

The present kernel measurements use powers of two in MPI rank count. They reach 16,384 ranks on Snellius, equivalent to 85.3 fully populated 192-core nodes. This brackets a 64-node test but does not demonstrate the requested class above 100 nodes. Exact node-aligned rank counts must also be tested; performance at 8,192 or 16,384 ranks should not be silently substituted for a 12,288-rank, 64-node measurement.

Class	Proposed nodes	MPI ranks	Present evidence
Small	1, 2, 4, 8, 10	192–1,920	Kernel curves bracket the range; application to 1,024 ranks
Medium	16, 32, 64	3,072–12,288	Kernel data at 4,096 and 8,192 ranks; 64 nodes bracketed
Large	128, 256	24,576–49,152	Not yet measured; requires a scaled problem and memory pilot

Table 1: Proposed node-aligned qualification points for a system with 192 CPU cores per node. Existing evidence and proposed measurements are deliberately separated.

For the small class, the current $L = 9$ three-dimensional and $L = 14$ two-dimensional problems can be retained as fixed-size strong-scaling tests. For 16–64 nodes, a three-dimensional $L = 10$ case should be added after a memory check. The 128- and 256-node points require a larger problem, such as an $L = 11$ octree, after a short pilot has established memory use and initialization cost. The purpose of the large case is to preserve enough cells per rank to measure the network at scale rather than to time an already saturated problem.

The representative two-phase case should accompany the kernels at small and medium scales. A large-scale application case should be promoted only after its domain size, interface area and output policy have been fixed so that each machine performs the same physical and numerical work.

5 Reproducibility and reporting

A procurement-quality result should record the following alongside every timing:

- the exact source revision and unmodified stock-test provenance;
- compiler, optimization flags, MPI implementation and launcher;
- node type, ranks per node, thread affinity and rank binding;
- mesh dimensionality, level, total cell count and cells per rank;
- wall time, MPI time, iteration count and any initialization excluded from the timed region; and
- at least three repeats after the initial qualification run, reporting the median and spread rather than only the best observation.

The current CSV files contain one observation for each configuration. They are sufficient to locate scaling regimes and design the next campaign, but they do not quantify system variability. Repeated node-aligned runs are therefore part of the proposed qualification procedure.

6 Conclusion

Basilisk provides a compact CPU benchmark with a direct connection to production multiphase-flow workloads. The stock Poisson and Laplacian kernels separate local computation from MPI

communication, while the verified two-phase case tests interface reconstruction, surface tension and the full projection step. Existing measurements cover the small class and bracket a 64-node, 192-core-per-node test. The next useful measurements are exact node-aligned repeats and scaled three-dimensional problems at 128 and 256 nodes. This distinction keeps present evidence separate from the capabilities that a future system must demonstrate.

References

- [1] S. Popinet, *Basilisk: MPI circle benchmark*, <https://basilisk.fr/src/test/mpi-circle.c>.
- [2] S. Popinet, *Basilisk: MPI Laplacian benchmark*, <https://basilisk.fr/src/test/mpi-laplacian.c>.
- [3] S. Popinet, An accurate adaptive solver for surface-tension-driven interfacial flows, *Journal of Computational Physics* **228** (2009), 5838–5866. doi:10.1016/j.jcp.2009.04.042.
- [4] M. O. Abu-Al-Saud, S. Popinet and H. A. Tchelepi, A conservative and well-balanced surface tension model, *Journal of Computational Physics* **371** (2018), 896–913. doi:10.1016/j.jcp.2018.02.022.